

HDOC - Day 11 Activity

C Programming Assignment

Menu-Driven Program Using switch-case

Problem Statement

Prepare the algorithm and test case table. Then write, compile, execute, and test a C program that accepts a positive integer and performs one of three digit-based operations according to the user's choice.

Operations:

1. Find the sum of squares of the digits using a while loop. Example: $123 \rightarrow 1^2 + 2^2 + 3^2 = 14$.
2. Find the sum of cubes of the digits using a for loop. Example: $123 \rightarrow 1^3 + 2^3 + 3^3 = 36$.
3. Find the difference between the sum of cubes and the sum of squares using a do-while loop. Example: $123 \rightarrow 36 - 14 = 22$.

Algorithm

1. START.
2. Input a positive integer n.
3. If $n \leq 0$, display an error message and STOP.
4. Display the menu:
 - 1 - Sum of squares of digits
 - 2 - Sum of cubes of digits
 - 3 - Difference (sum of cubes - sum of squares)
5. Input choice.
6. Use switch(choice):
Case 1:
 - a. Set temp = n and sumSq = 0.

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b. WHILE temp > 0:
digit = temp % 10
sumSq = sumSq + digit * digit
temp = temp / 10
c. Display sumSq.
Case 2:
a. Set sumCube = 0.
b. FOR temp = n; temp > 0; temp = temp / 10:
digit = temp % 10
sumCube = sumCube + digit * digit * digit
c. Display sumCube.
Case 3:
a. Set temp = n, sumSq = 0 and sumCube = 0.
b. DO:
digit = temp % 10
sumSq = sumSq + digit * digit
sumCube = sumCube + digit * digit * digit
temp = temp / 10
WHILE temp > 0
c. Display sumCube - sumSq.
Default: Display "Invalid choice".
7. STOP

```

Test Case Table

The following test cases check all three valid menu choices as well as invalid input conditions.

Case	Input	Choice	Expected output	Purpose
1	123	1	Sum of squares = 14	Valid choice 1
2	123	2	Sum of cubes = 36	Valid choice 2
3	123	3	Difference = 22	Valid choice 3
4	45	1	Sum of squares = 41	Two-digit input
5	7	2	Sum of cubes = 343	Single-digit input
6	0	1	Error: Enter a positive integer	Invalid number
7	123	4	Invalid choice	Invalid menu choice

C Program

```

#include <stdio.h>
int main() {
int n, choice, temp, digit;
int sumSq = 0, sumCube = 0;

```

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printf("Enter a positive integer: ");
scanf("%d", &n);
if (n <= 0) {
printf("Error: enter a positive integer.\n");
return 0;
}
printf("\n1. Sum of squares of digits\n");
printf("2. Sum of cubes of digits\n");
printf("3. Difference (cubes - squares)\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
temp = n;
while (temp > 0) {
digit = temp % 10;
sumSq += digit * digit;
temp /= 10;
}
printf("Sum of squares = %d\n", sumSq);
break;
case 2:
for (temp = n; temp > 0; temp /= 10) {
digit = temp % 10;
sumCube += digit * digit * digit;
}
printf("Sum of cubes = %d\n", sumCube);
break;
case 3:
temp = n;
do {
digit = temp % 10;
sumSq += digit * digit;
sumCube += digit * digit * digit;
temp /= 10;
} while (temp > 0);
printf("Difference = %d\n", sumCube - sumSq);
break;
default:
printf("Invalid choice.\n");
}
```

```
return 0;
}
```

Compilation, Execution and Testing

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ cc menu.c
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
Enter your choice: 1
Sum of squares = 14
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
Enter your choice: 2
Sum of cubes = 36
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
Enter your choice: 3
Difference = 22
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 45
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
Enter your choice: 1
Sum of squares = 41
```

```
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 7
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
Enter your choice: 2
Sum of cubes = 343
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 0
Error: enter a positive integer.
(kali㉿kali)-[~]
$ ./a.out
Enter a positive integer: 123
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (cubes - squares)
Enter your choice: 4
Invalid choice.
(kali㉿kali)-[~]
$
```

Result: The menu-driven C program was written using switch-case. The required operations were implemented with a while loop, for loop, and do-while loop. The program gives the expected results for the prepared valid test cases and also handles a non-positive integer and an invalid menu choice.